Supporting file for reproducing results of the article: Lanthanide molecular nanomagnets as probabilistic bits

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This document offers a help for reproducing the graphics. As we comment in the 'readme.txt', the following modules need to be installed:

1. Pandas
2. Numpy
3. Scipy
4. Math
5. Random
6. Time
7. Matplotlib

It is important to emphasize that the file named 'main.py' contains the body of the simulator; the user could only change the configuration section at the beginning of the script (lines 20-38).

Lanthanide-based, molecular, isolated spin p-bits at constant field

As an example, we present the experiment at 80K employing $N = 10^5$ spins. In the case of 80, 40, and 2K, the program compares the results of the magnetic moment with the experimental data extracted from F.-S. Guo, B. M. Day, Y.-C. Chen, M.-L. Tong, A. Mansikkamäki and R. A. Layfield, Science, 2018, 362, 1400–1403.

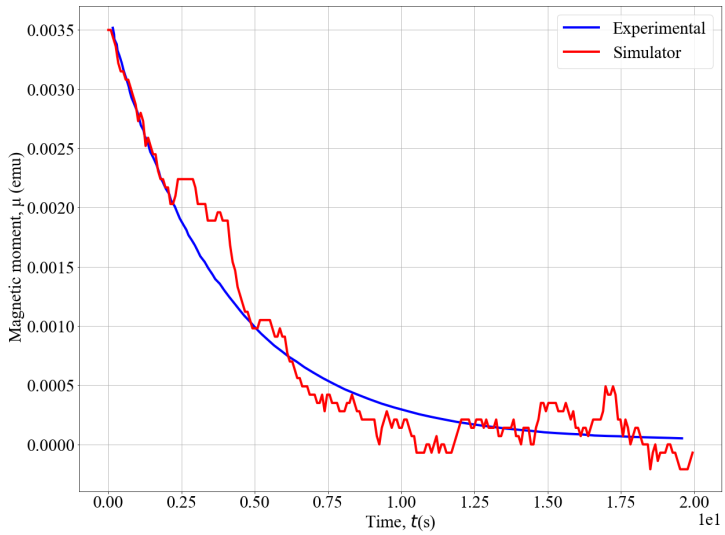

Figure 1: Relaxation plot for 100 spins at 80K.

Table 1: Variable's configurations at 80K, 10^5 spins, for $[(Cp^{iPr5})Dy(Cp^*)]^+$ (Cp^{iPr5}, penta-iso-propylcyclopentadienyl; Cp^*, pentamethylcyclopentadienyl).

Variable	@80K
option	0
option 2spin	0
N ex	100000
flag	23
T	80
B	0
B max	NC
B pbit2	NC
save	1
cycles	NC
starting mode	0
repicable exp	1
tau exp	50
total time	20
probability	1.9998

Lanthanide-based, molecular, dynamically driven spin p-bits

In this case, we need a more delicate discretization, thinking about extrapolating the exponential factor from 50 to 5000, the total probability for the spin to flip will be 0.019998%. In the file named "ac fitting" we present the ac-calculation for multiple magnetic fields.

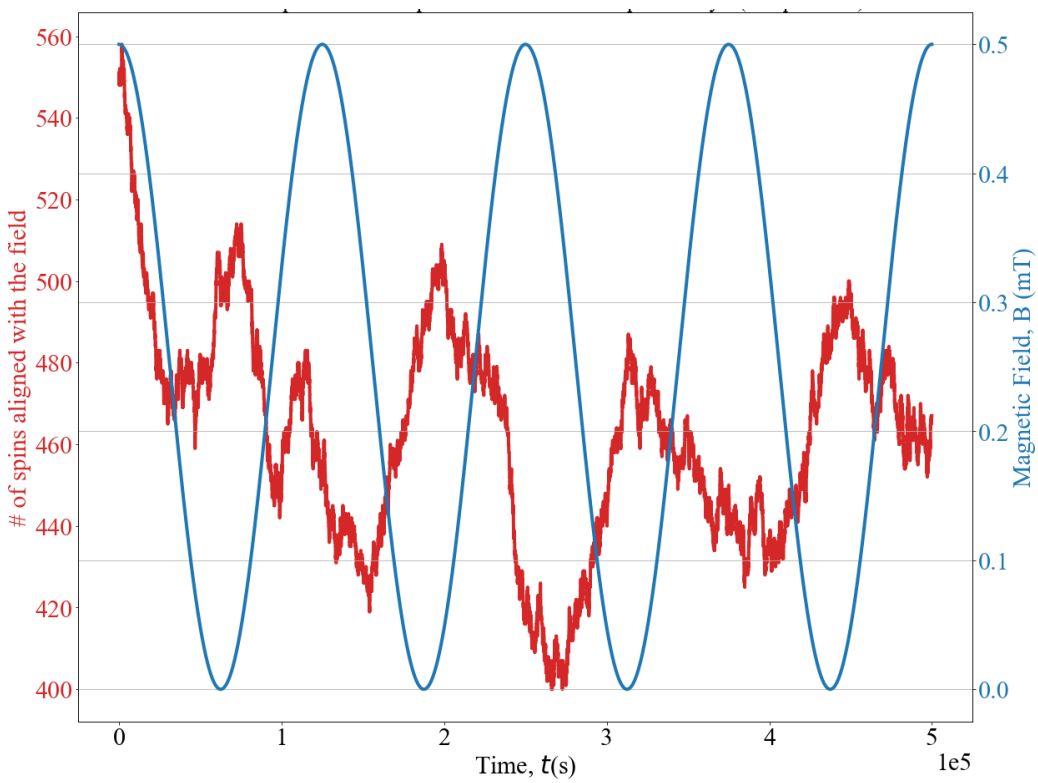

Figure 2: Relaxation plot for 1000 spins at 20mK.

Variable	@20mK
option	1
option 2spin	0
N ex	1000
flag	23
T	20/1000
B	NC
B max	0.00025
B pbit2	NC
save	1
cycles	4
starting mode	1-0.55
repicable exp	0
tau exp	5000
total time	$5 * 10^4$
probability	0.019998

Table 2: Variable's configurations at 20mK, 1000 spins.

Section 3.3. Lanthanide-based, molecular spin p-bit network

This part of the simulator follows the calculations of two p-bit network. To reproduce the result, we present the next table with all the necessary values.

Variable	@40K
option	0
option 2spin	1
N ex	NC
flag	23
T	40
B	0.01
B max	NC
B pbit2	0.2
save	1
cycles	NC
starting mode	0
repicable exp	0
tau exp	50
total time	$35 * 10^6$
probability	1.9998

Table 3: Variable's configurations at 40K to analyze the effect of a p-bit to another, for $[(Cp^{iPr5})Dy(Cp^*)]^+$ (Cp^{iPr5}, penta-iso-propylcyclopentadienyl; Cp^*, pentamethylcyclopentadienyl).